
EE 504: Switch Mode Power Conversion

Course Code	EE 504
Course Name	Switch Mode Power Conversion
L-T-P-C	2.5-0.5-0-3
Prerequisites	EE 309 Power Electronics and Teachers Consent
Intended for	UG/MS/PhD
Distribution	Elective
Approval	5th Senate (Oct 2013)

Course Contents

- **Introduction:** About Switch Mode Power Conversion; overview of the course; industrial relevance of this topic; SMPC requirements. [2 Lectures]
- **Basic DC to DC power converters:** Basic DC-DC converters - buck, boost, buck-boost & Cuk converters - and their principles of operation; continuous and discontinuous modes of operation; SEPIC converter. [6 Lectures]
- **Power semiconductor switches:** Review of power diodes, Schottky diodes, power MOSFETs & IGBTs. Recent developments in power devices for switch mode power supplies. Drive requirements, switching performance and snubber design. Selection of devices & basic heat sink design. [6 Lectures]
- **Transformer-Isolated Converters:** Single-switch and multi-switch transformer-isolated DC-DC converters. Flyback and forward converters; transformer isolated half-bridge, full-bridge converters. Push-pull converters. Voltage-fed and current-fed converters. [6 Lectures]
- **Magnetic Component Design:** Magnetic core materials and performance; basic inductor and transformer design; practical magnetic design; design aspects to be considered for designing transformers for specific applications flyback, push-pull, bridge, forward converters. [4 Lectures]
- **Switching Regulator Control:** Small-signal models for switching regulators. Performance analysis and design of closed-loop system under different control methods, and operating modes. Measurement of small signal transfer functions. [6 Lectures]
- **Soft-Switched and Resonant DC-DC Power Converters:** Motivation. Hard-switching vs soft-switching. Introduction to resonant power converters and their characteristics. Detailed study of a few soft-transition converters. [6 Lectures]
- **Single-Phase Power-Factor Correction:** Problems due to harmonics in the current drawn by equipment. Basic concept of active power-factor correction (PFC) techniques. Performance analysis and comparison of different PFC techniques. [6 Lectures]

Reference Material

1. N Mohan, T M Undel and W P Robbins, **Power Electronics: Converters, Applications and Design**, Wiley
2. A I Pressman, **Switching Power Supply Design**, McGraw-Hill
3. Selected Conference and Journal Articles
4. **Texas Instruments Power Management Application Notes** including articles from the well known Unit rode Seminar Series.
5. **Application Notes** from International Rectifiers and other Power Devices and ICs manufacturers.